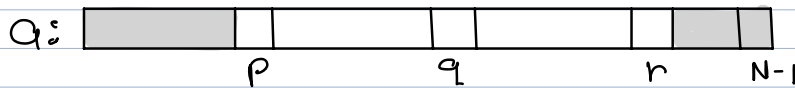


void merge(int * a, int p, int q, int r).



Pre-condition-1: $0 \leq p \leq q < r < N$.

Pre-Condition-2: $a[p \dots q]$ and $a[q+1 \dots r]$

Post-condition: ^{Sorted.} $a[p \dots r]$ is sorted.

Let a and b be two integers such that $a \leq b$.

All integers n , $a \leq n \leq b$. $[a, b]$ closed interval

$a \leq n < b$. $[a, b)$ Semi-open

$a < n \leq b$ $(a, b]$ Semi-open.

$a < n < b$. (a, b) open interval.

0 1 2 3 4 5 6 7 8 9 10
 $2 \leq n \leq 7$

0 1 2 3 4 5 6 7 8 9 10
 $2 \leq n < 7$

0 1 2 3 4 5 6 7 8 9 10
 $2 < n \leq 7$

0 1 2 3 4 5 6 7 8 9 10
 $2 < n < 7$.

Find out number of intervals in

$$[a, b], (a, b), [a, b), (a, b] \quad 9-5$$

$$(5, 9] \quad 9-5$$

$$(a, b] \quad b-a.$$

$$[a, b) \quad b-a$$

$$[a, b] \quad b-a+1.$$

$$(a, b) \quad b-a-1.$$

Let $a, b \in \mathbb{R}$. Find out number of integers between $[a, b]$ (a, b) $[a, b)$ $(a, b]$.

$$[a] \lceil a \rceil \lfloor b \rfloor \lceil b \rceil \quad \textcircled{2,3} \mid 50$$



Let $x \in \mathbb{R}^+$ and $n \in \mathbb{N}$.

$$\lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor + \lfloor x + \frac{1}{3} \rfloor + \dots + \lfloor x + \frac{1}{n} \rfloor = \lfloor nx \rfloor$$

$$\lfloor x \rfloor + \sum_{2 \leq k \leq n} \lfloor x + \frac{1}{k} \rfloor = \lfloor nx \rfloor$$

$$x \begin{matrix} \swarrow \\ \lfloor x \rfloor \end{matrix} \searrow \{x\}$$

$$\lfloor \{x\} \rfloor = 0$$

$$\lceil \{x\} \rceil = 1$$

$$x = \lfloor x \rfloor + \{x\} \quad \text{M.M.2}$$

Sub-array-1: $a[p \dots q]$ $[p, q]$ $q-p+1$

Sub-array-2: $a[q+1 \dots r]$ $(q, r]$ $r-q$

$$0 \leq p \leq q.$$

$$q < r.$$

$$p \leq q$$

$$0 < r-q$$

$$p-p \leq q-p.$$

$$r-q > 0$$

$$0 \leq q-p$$

$$\text{len}(\text{SA.2}) > 0$$

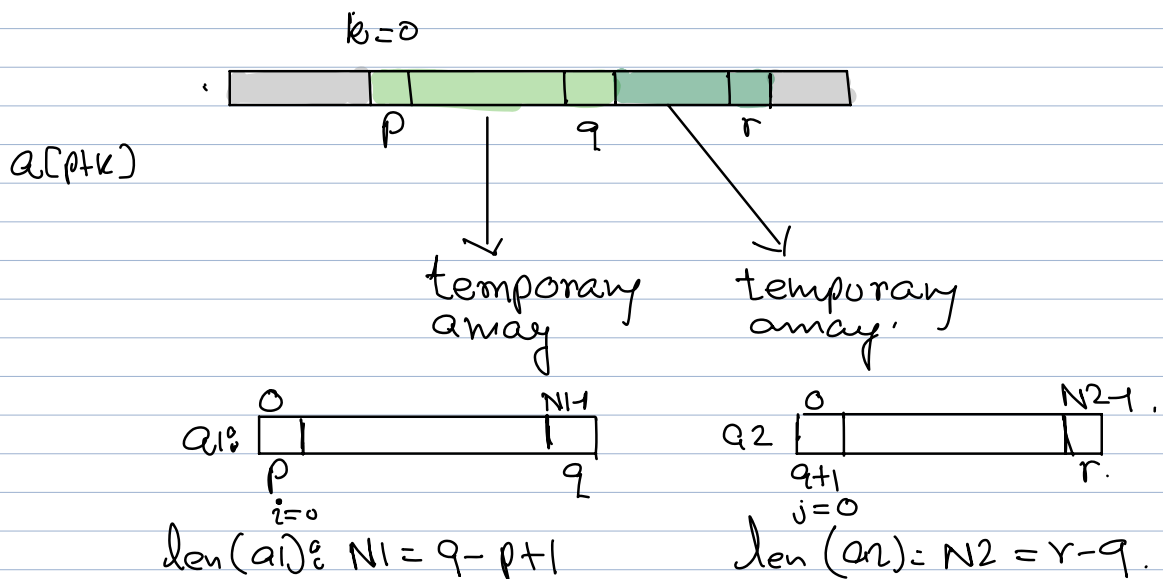
$$1 \leq q-p+1$$

$\therefore \text{SA.2}$ is not empty.

$$q-p+1 \geq 1$$

$$q-p+1 > 0$$

$\therefore \text{SA.1}$ is Non-empty



$$a_1[0 \dots N_1-1] \leftarrow a[p \dots q]$$

$$a_2[0 \dots N_2-1] \leftarrow a[q+1 \dots r]$$

$$i=0, j=0, k=0$$

```

while ( — )
{
    if ( $a_1[i] \leq a_2[j]$ )
    {
         $a[p+k] = a_1[i];$ 
         $k++;$   $i++;$ 
    }
    else
    {
         $a[p+k] = a_2[j];$ 

```

```

    }
    k++; j++;
}

```

```

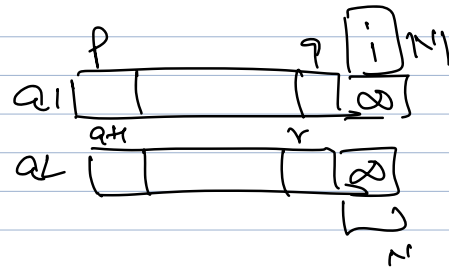
i = 0;
j = 0;
k = 0;
cnt = 0;
N = r - p + 1;
while (cnt < N)

```

```

{
    if (a1[i] <= a2[j])
    {
        a[p+k] = a1[i];
        i++;
        k++;
    }
    else
    {
        a[p+k] = a2[j];
        j++;
        k++;
    }
}

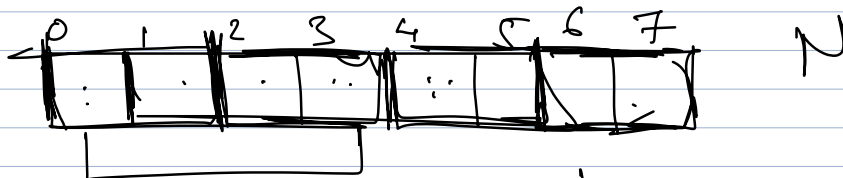
```



```

    cnt++;
}

```



$\text{merge}(a, 0, 0, 1)$
 $p \quad q \quad r$
 $\leq <$

$\text{merge}(a, 2, 2, 3)$
 $p \quad q \quad r$

$\text{merge}(a, 0, 1, 3)$
 $p \quad q \quad r$

$\text{merge}(a, 4, 4, 5)$
 $p \quad q \quad r$

$\text{merge}(a, 6, 6, 7)$
 $p \quad q \quad r$

$\text{merge}(a, 4, 5, 7)$
 $p \quad q \quad r$

$\text{merge}(a, 0, 3, 7)$

```

void merge_sort (int* a, int p, int r)
{
    int q = (p+r)/2;
    if (p < r)
    {
        merge_sort (a, p, q);
        merge_sort (a, q+1, r);
        merge (a, p, q, r);
    }
}

```

$n. f(n-1)$ if $n > 0$

$f(n) =$

1

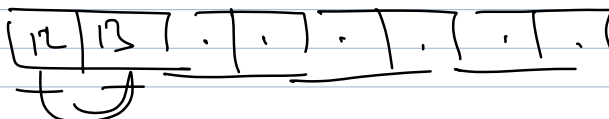
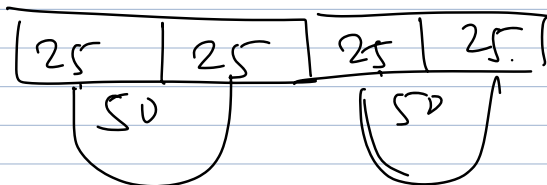
if $n = 0.$

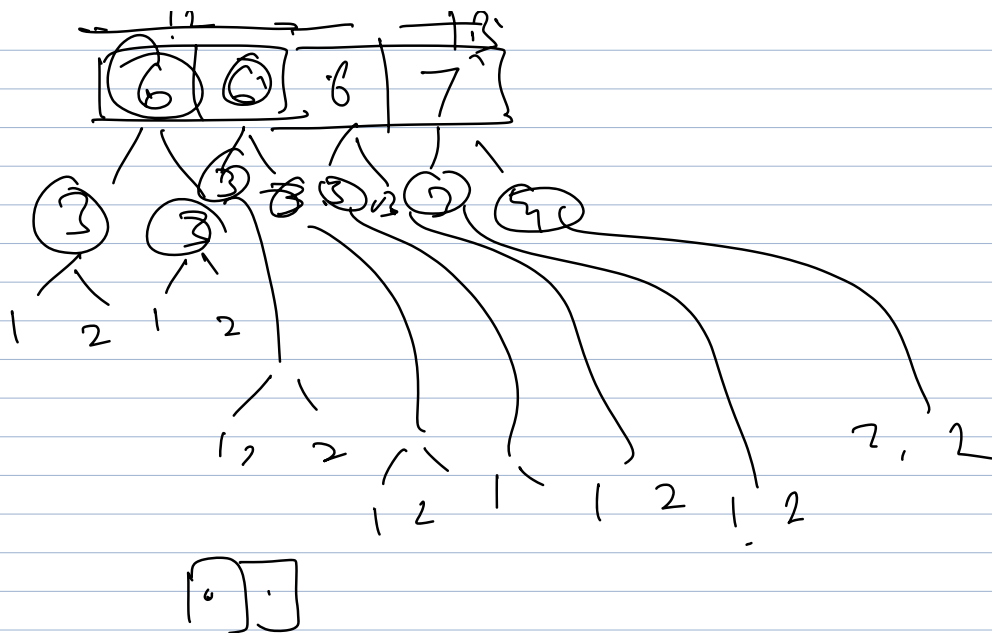


$(0, q) \rightarrow \text{sort}$

$(q+1, N-1) \rightarrow \text{sort}$

$\text{merge}(a, p, q, r)$





$$q = \lfloor \frac{(p+r)}{2} \rfloor$$

M.S. (a, p, q)

M.S. (a, q+1, r)

M(a, p, q, r) -- if $p < r$.

MS (a, p, r) =

return if $p \geq r$.

$N=11$

M.S. (a, 0, 10)

$p=0, r=10, q=5$

M.S. (a, 0, 5)

$p=0, r=5, q=2$

M.S. (a, 6, 10)

M.S. (a, 0, 2)

$p=0, r=2, q=1$

M.S. (a, 3, 5)

$p=3, r=5, q=4$

M.S. (a, 0, 1)

$p=0, r=1, q=0$

M.S. (a, 2, 2)

$p=2, r=2$

return

M.S. (a, 3, 4)

$p=3, r=4, q=3$

M.S. (a, 5, 5)

$p=5, r=5$

return

M.S. (a, 0, 0)

$p=0, r=0$

return

M.S. (a, 1, 1)

$p=1, r=1$

return

M.S. (a, 3, 3)

$p=3, r=3$

return

M.S. (a, 4, 4)

$p=4, r=4$

return

① M(a, 0, 0, 1)

② M(a, 0, 1, 2)

③ Merge(a, 3, 3, 4)

④ M(a, 3, 4, 5)

⑤ M(a, 0, 2, 5)